

Adel Iqbal

School:
6360 Main Street
Houston, TX 77005
(210) 889-7709, ai19@rice.edu

Home:
312 Tularosa Ridge
Helotes, TX 78023

Education

Rice University - Houston, TX
Bachelor of Arts, May 2024
Major: Neuroscience

Research Experience

Baylor College of Medicine (Houston, TX, USA)
Basic Sciences Research Assistant

- Led independent research project investigating age-related changes in glial cell-surface protein expression using *Drosophila* as a model organism.
- Performed comprehensive transcriptomic analysis, identifying and validating candidate genes through single-cell RNA sequencing data.
- Designed and executed genetic screens using RNAi knockouts and Gal4/UAS system to assess candidate genes' effects on lifespan and locomotion.
- Applied multiple technical approaches including bioinformatics analysis, confocal microscopy imaging, and systematic data quality control to identify contamination issues.

El Bari Community Health Center (San Antonio, TX, USA)
Basic Sciences Research Assistant

- Designed and led qualitative research exploring barriers to childhood immunizations through 13 focus groups with parents of unvaccinated children.
- Developed standardized questionnaires and conducted thematic analysis of participant responses, identifying key barriers including substance abuse, lack of motivation, and limited mobility.
- Gained comprehensive experience in qualitative research methodology, including participant recruitment, data coding, and theme analysis.
- Applied findings to inform future goals of conducting public health research on preventive care access and developing community-based solutions.

University of Texas Health Science Center at San Antonio (San Antonio, TX, USA)
Basic Sciences Research Assistant

- Investigated the functional role of TMEM94 in mitochondrial metabolism through systematic pharmacological screening of metabolic modulators.
 - Performed live-cell imaging and subcellular fractionation to characterize TMEM94 localization and protein-protein interactions.
 - Conducted high-throughput drug screening assays to assess metabolic pathway perturbations, utilizing fluorescence microscopy and metabolic indicators.
 - Assisted in optimization of isolation protocols for mitochondrial preparations and western blot analysis of TMEM94 expression patterns.
-

Healthcare Experience

Baylor College of Medicine; Texas Children's Hospital (Houston, TX, USA)

Clinical Research Assistant

- Coordinated multi-site case-control study (n=17) on responsive neurostimulation and memory, managing IRB documentation and patient recruitment.
- Developed standardized protocols for cognitive assessments and neurostimulation parameters, analyzing outcomes with MATLAB and SPSS.
- Demonstrated significant improvement in delayed verbal recall ($p=0.04$, $d=0.627$) via dcs
- Presented findings at academic symposiums and contributed to published manuscript

Bexar County ESD No. 8 Fire Department (San Antonio, TX, USA)

Emergency Medical Technician

- Performed comprehensive patient assessments, administered basic life support interventions, and provided emergency medical care according to established protocols.
- Collaborated with paramedics, firefighters, and hospital staff to ensure seamless patient care and efficient emergency response coordination.
- Maintained accurate patient care reports and medical documentation while ensuring compliance with HIPAA regulations and departmental standards.

Serenity Hospice Care (San Antonio, TX, USA)

Hospice Care Volunteer

- Provided weekly companionship and emotional support to terminally ill patients during 4-hour shifts, developing strong active listening skills and cultural sensitivity.
- Assisted nursing staff with non-medical comfort measures and patient care tasks while ensuring dignity and compassionate care delivery.
- Supported families through difficult transitions by offering emotional support and respite care, demonstrating empathy and emotional resilience.
- Collaborated with an interdisciplinary healthcare team to ensure comprehensive patient support, gaining valuable exposure to end-of-life care principles.

Community Service

Academic Fellow (Houston, TX, USA)

Member

- Selected for prestigious Sid Richardson College Academic Fellows Society at Rice University, recognizing outstanding academic achievement and commitment to peer mentorship.
- Provided comprehensive one-on-one tutoring and led group review sessions across multiple subjects, demonstrating strong teaching and communication skills.
- Maintained regular office hours and organized targeted study sessions to ensure accessible academic support for residential college community.
- Combined academic excellence with dedicated service through peer education, enhancing learning outcomes for fellow students.

San Antonio Food Bank (San Antonio, TX, USA)

Food bank volunteer

- Coordinated weekly food distribution efforts with team of 10 volunteers at San Antonio Food Bank, serving 200+ families and elderly residents.
- Provided essential nutrition and diabetes management education while distributing meals to underserved community members.
- Demonstrated strong interpersonal skills through effective communication with diverse population groups and volunteer team coordination.
- Developed deeper understanding of food insecurity and social determinants of health impacting local community outcomes.

Leadership

Letters for Change (Riyadh, Saudi Arabia)

Co-Founder

- Founded and led major initiatives at L4C, developing bilingual curricula and English proficiency programs reaching 600+ disadvantaged Saudi children.
- Managed team of 12 educators to create 35+ adaptable curricula across multiple competency levels, implementing programs through structured 9-week sessions.
- Coordinated Rice University's inaugural International Education Week while continuing international education advocacy efforts.
- Conducted comprehensive research using SPSS to assess medical proficiency in Saudi public schools, presenting findings to Ministry of Education to advocate for school nursing programs.

Rice Neuroscience Society NeuroCamp (Houston, TX, USA)

President & Co-Director

- Led 18-member committee as Executive Vice President of NeuroCamp, developing and implementing free STEM summer program for underserved Texas high school students.
- Expanded program reach to 600+ students through strategic community partnerships, focusing on underrepresented populations in STEM fields.
- Achieved exceptional program outcomes with 95% participant satisfaction (4/5 or higher) and 82% reporting increased interest in STEM careers.
- Created lasting impact through targeted scholarship guidance and college preparation resources, directly influencing students' academic and career trajectories.

Rice University BioSciences Department (Houston, TX, USA)

Lead Teaching Assistant

- Led team of 7 teaching assistants for introductory biology course, organizing biweekly discussions for 45 students while managing assignments and grading responsibilities.
- Created innovative teaching materials, including simplified diagrams and targeted practice problems, to improve student comprehension of challenging concepts like cellular respiration.
- Maintained regular office hours to provide individualized support and course material revision, demonstrating commitment to student success.
- Developed adaptable teaching methods to accommodate diverse learning styles, gaining valuable experience in making complex information accessible for future patient education.

Extracurricular Activities and Leisure

Aerobic exercise (running, biking, rowing)

Piano (actively learning)

Language learning (arabic, spanish, turkish, urdu)